Phase 1: Data Preparation and Baseline Summarizer

Project: Multilingual Courtroom Judgment Simplifier

Objective: To prepare the dataset and build a baseline system that explains a court judgment in plain language, so that later phases can be measured against it.

1. Dataset

- The dataset has 2000 synthetic Indian civil judgments in CSV format. Real judgments are confidential and hard to access, so synthetic cases were generated using an LLM.
- Each case has 7 fields: Case_Name, Facts_of_the_Case, Issues, Arguments, Ratio_Decidendi, Final_Order and Legal_Terms.
- Quality checks found no missing values and no duplicate rows.
- A full case is about 525 words (430 to 705), so it fits in one model prompt and needs no chunking.
- Each case lists 3 to 4 legal terms with how they were used. There are 31 unique terms across the dataset (for example Bona Fide, Estoppel, Prima Facie, Res Judicata). These will be the basis of the legal-term glossary.

Table 1: Length of each field (words)

Field	Min	Median	Max
Facts	105	129	178
Issues	33	51	67
Arguments	66	119	182
Ratio decidendi	82	147	219
Final order	30	49	86

2. Preprocessing

- Extra spaces were removed and the text was cleaned.
- The bracketed header at the start of the facts was split into three new fields: court, case number and judgment date. This worked for all 2000 cases.
- A combined input text was built from the facts, issues, arguments and final order. The ratio decidendi was kept out of the input and reserved as the reference answer for Phase 2.
- The data was split into training (1600), validation (200) and test (200) sets, using a fixed random seed so the split can be repeated. All files were saved to Google Drive.

3. Baseline summarizer

- Model: Qwen2.5-3B-Instruct, run on a Google Colab T4 GPU.
- Method: a prompt asking the model to explain the judgment in 5 short, simple sentences using only facts from the text, with no legal jargon. Generation was deterministic so results can be repeated.
- Run: 20 test cases were summarized, and each result was saved to outputs/baseline_summaries.jsonl.

4. Evaluation

Four summaries were manually compared with the actual final order. The checks were: correct outcome, missing details, wrong or invented statements, and plain language.

Table 2: Manual review of baseline summaries

Case	Outcome	Main problems
Meghdoot Agro vs. Padmavathi Gowda	Partly correct	Missed easement of necessity vs prescriptive easement; lost the 8-foot detail; costs order misread
Anantham Exports v. Adarsh Gruh Nirman	Partly correct	Ground for eviction missing; "arrears accepted" is inaccurate; "part of the land" is imprecise
Kamala Deshmukh vs. Trilok Infratech	Correct	Wrong party said to pay the balance; amount omitted
Suresh Pathan v. Iqbal Fernandes	Correct	Forfeiture of share missing; costs order misread

Patterns found

1. "No order as to costs" was misread in 2 of 4 cases.
2. Legal distinctions and specific figures (the 8-foot width, Rs. 84,60,000) were lost.
3. Roles of the parties can get mixed up.
4. The language is simple and readable, but some summaries list findings without saying who won.

5. Conclusion

The baseline produces readable plain-language summaries and usually gets the general outcome right. It still loses legal distinctions, misstates some orders and has no check against the source text. This supports the next steps of the project: ratio decidendi extraction, grounded legal-term explanations using retrieval, and a hallucination guard.

6. Limitations

- The data is synthetic and uniform, so results may look better than on real judgments.
- Only 4 of the 20 summaries were reviewed by hand, and only against the final order.
- The dataset has no reference summaries, so summaries were judged by reading, not by a numeric score.

7. Deliverables

- Cleaned dataset and splits: clean.csv, train.csv, val.csv, test.csv
- Notebook: 01_phase1_and_2_cleaning_and_baseline.ipynb (Phase 1 cells)
- 20 baseline summaries: outputs/baseline_summaries.jsonl